Research Summary
NAME: Emile Rugamika Chimusa

eRA COMMONS USER NAME (credential, e.g., agency login): EMILECHIMUSA

POSITION TITLE: Professor of Bioinformatics and Computational Biology – Northumbria University, UK

EDUCATION/TRAINING

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
Sacred Heart High School College, Kisangani	BOTH	07/1997	Mathematics and Physics
University of Kinshasa, Kinshasa	BSc	07/2005	Applied Mathematics
African Institute for Mathematical Sciences (AIMS), University of Western Cape, Cape Town	MSc	06/2008	Mathematical Sciences
University of Cape Town, Cape Town, Western Cape	PhD	06/2013	Bioinformatics and Computational Biology

A. Personal Statement

My expertise and research build on my bioinformatics background and focus on data science and AI methods and tools development pertinent to human genomic diversity to uncover the role of genetics and environment in determining the risk of and disease susceptibility. I have long-standing experience in developing software packages (https://github.com/echimusa) and implementing workflows for Omics data analysis and FAIR principles to facilitate reproducible and impactful science. I have a record of peer-reviewed research publications, and educational pedagogy articles (ORCID:0000-0001-8846-2047). For the last 2 years, I served as a member or chair of a steering group for Race Equality, Diversity, and Inclusion at Northumbria University.

I have been awarded The Academy of Medical Sciences Professorship (APR9\1024, UK) to investigate novel artificial intelligence approaches for disease risk prediction and stratification. I have been an expert in protocol design and strategy development in several African Bioscience projects funded by the Bill & Melinda Gates Foundation. For the past 3 years, I have been a European Commission expert/Reviewer and reviewed several Horizon Europe grant applications and recently joined the Pool of Experts and Follow on Fund Committee. Recently nominated as Deputy of Steering Group of Northeast Fatigue Research Network. I accumulated a record of Scientific or guest editor in several international Bioscience related journals. I have organized various workshops and co-organized events on Big Data, Analytics, and Machine Intelligence and spoken at International events on these topics.

1. **Chimusa ER***, Mbiyavanga M, Mazandu GK, Mulder NJ. ancGWAS: A post genome-wide association study method for interaction, pathway and ancestry analysis in homogeneous and admixed populations. Bioinformatics. 2016 Feb 15;32(4):549-56. PMID: 26508762
2. **Chimusa ER***, Mbiyavanga M, Masilela V, Kumuthini J (2015) “Broadband” Bioinformatics Skills Transfer with the Knowledge Transfer Programme (KTP): Educational Model for Upliftment and Sustainable Development. PLoS Comput Biol 11(11): e1004512. PMID: 26583922
3. Mugo JW, Geza E, Defo J, Elsheikh SSM, Mazandu GK, Mulder NJ, **Chimusa ER*.** A multi-scenario genome-wide medical population genetics simulation framework. Bioinformatics. 2017 Oct 1;33(19):2995-3002. PMID: 28957497
4. **Chimusa ER***, Zaitlen N, Daya M, Möller M, van Helden PD, Mulder NJ, Price AL, Hoal EG. Genome-wide association study of ancestry-specific TB risk in the South African Coloured population. Hum Mol Genet. 2014 Feb 1;23(3):796-809. PMID: 24057671

Ongoing and recently completed projects that I would like to highlight include:

Ongoing Research Support

Academy of Medical Sciences Professorship (APR9/1024), UK
Chimusa (PI)
03/05/2023–03/05/2027
Optimal framework for disease risk prediction and stratification
Goal: Develop next-generation of ancestrally and environmentally sensitive disease risk prediction.

Past Research Support

NIH (1 U2R TW012131-01)
Chimusa (PI)
03/31/2017–03/30/2020
Computational Omics and Biomedical Informatics Program (COBIP)
Goal: Develop a computational omics and Biomedical informatics program (COBIP) to train African biomedical data scientists and faculty in data science, computational omics, clinical informatics, and imaging data science.

NIH (U24HL135600-01)
Wonkam (PI)
03/31/2017–03/30/2020
Developing a Sickle Africa Data Coordinating Center (SADaCC)
Goal: The major goal of this is to establish a Data Coordinating Center (DCC) to be called the Sickle Africa Data Coordinating Center (SADaCC), which will allow us to build a robust platform to support the activities of a companion Sickle Cell Disease Network in Africa.
Role: Expert/Consultant

NIH/NHGRI (1U54HG009790-01)
Wonkam Ambroise (PI)
07/01/2017–06/30/2022
Individual Findings in Genetic Research in Africa (IFGeneRA)
Goal: The major goal is to develop a progressive evidence base series of research activities that will inform the development of context- and country-specific policies relating to the return of individual genetic research results, including Incidental Findings, in genomics research in Africa.
Role: Co-Investigator

Wellcome Trust/AESA (H3A/18/001)
Wonkam (PI)
07/01/2017–06/30/2022
Hearing Impairment Genetics Studies in Africa (HI-GENES Africa)
Goal: The major goal is to complement the 1U01HG009716-01 grant in order to: 1) use a mouse model for functional studies of novel genes identified by Whole genome sequencing (WES), in families from Cameroon, Mali, Ghana, and South Africa with prelingual NSHI, 2) to implement communities engagement activities with families, 3) and to implement a fellowship program for the training of post-graduate students in and Human genetics and genetic counseling.
Role: Co-Investigator

NIH/NHGRI (U01HG009716)
Wonkam Ambroise (PI)
07/01/2017 – 06/30/2022
Hearing Impairment Genetics Studies in Africa (HI-GENES Africa)
Goal: The major goal is to use Whole genome sequencing (WES), to study the largest sample of sub-Saharan Africans from Cameroon, Mali, Ghana, and South Africa with prelingual NSHI, in order to identify novel NSHI genes and to better understand the genetic etiology of NSHI in African populations.
Role: Co-Investigator

NIH/NHGRI H3Africa Program (U24HG006941)

Mulder Nicola (PI)
08/2012–07/2022
H3ABioNet: A sustainable African Bioinformatics Network for H3Africa (H3ABioNet)
Goal: To develop a Pan-African bioinformatics network to support H3Africa projects
Role: Co-Investigator

AIMS/IDRC (AASRG/SAMCD2008016S)
Chimusa (PI)
04/15/2018–12/28/2020
Computational method to understand population admixtures
Goal: To design a novel local ancestry inference approach for multi-ways admixture populations.

NRF (South Africa/Tunisia/762988)
Dandara (PI)
01/04/2019-30/03/2022
Molecular profiling of breast cancer for markers of genetic susceptibility and treatment response in Tunisian and South African populations
Goal: The aim of this project is to characterize molecular variants involved in breast cancer susceptibility and tamoxifen treatment response among Tunisian and South African breast cancer patients
Role: Co-Investigator

B. Positions, Scientific Appointments, and Honors

Positions and Scientific Appointments

2022 - present	Professor of Bioinformatics & Computational Biology, Department of Applied Science, FHL, Northumbria University Newcastle, UK
2019-2022	Associate Professor, University of Cape Town, Department of Pathology, Division of Human Genetics, Cape Town, South Africa
2016-2022	Convener of the Medical Sciences BSc Honours program in Human Genetics and Forensic Genetics at the Faculty of Health Sciences, University of Cape Town, Cape Town South Africa
2016-2019	Senior Lecturer, University of Cape Town, Department of Pathology, Division of Human Genetics, Cape Town, South Africa
2014-2015	Lecturer/researcher, University of Cape Town, Cape Town, South Africa
2013-2014	Biostatistical Scientific Researcher, Centre for Proteomic and Genomic research (CPGR)., Cape Town
2013-2014	Postdoctoral fellowship, University of Cape Town, Cape Town, South Africa
2011-2012	IT and Teaching assistant, African Institute for mathematical Sciences (AIMS), Cape Town
2005-2007	PostgreSQL database administrator, Department of Information Technology/United Nations Mission DR Congo (MONUC) Independent Electoral Commission (CEI), Kinshasa

Other Experience and Professional Memberships

2019-present	Member of Editorial Board, Frontiers in Oncology
2018-present	Member, Institute for Infectious Disease and Molecular Medicine
2017-present	Associate Editorial Board Member, Bentham Science
2014-present	Member and Public relation, South African Society for Bioinformatics (SASBi)
2013-present	Member, Human Heredity and Health in Africa (H3Africa)
2011-present	Member, Southern African Society for Human Genetics (SASHG)
2010-present	Member, International Society for Computational Biology (ISCB)

Honors

2024	Senior Fellowship, UK Professional Standards Framework for teaching and learning support in higher education.
2018	Elected as Future Leader, Science, and Technology in Society (STS) forum
2017	Elected as an Affiliate of The World Academy of Sciences (TWAS)

2017	Awarded C2 Rating, National Research Foundation
2014	Summer school in Advanced Monte Carlo Methods for Complex Inference Problems., Isaac Newton Institute for Mathematical Sciences, Cambridge, UK
2012-2014	Carnegie Corporation of New York, Ph.D. Scholarship and Postdoctoral award, University of Cape Town
2011	Young Investigator Awards, 12th International Congress of Human Genetics and The American Society of Human Genetics, Young Investigator Awards
2010	Prize, Intel Software Multi-core and Parallel programming, Stellenbosch University

C. Contributions to Science

My contributions to science have been in the field of computational statistical genetics and population genetics, where I have designed and applied computational methods for identifying genetic associations and deriving risk prediction in population-scale datasets, dissecting the genetics of diseases and traits in population cohorts.

1. Genetic Epidemiology and Genome wide association studies

Large genome wide association studies (GWAS) have been crucial in detecting disease associated variants in recent years. I have contributed to the field by both developing novel methods as well as performing GWAS in African structured and admixed populations.

a. Awany D, Allali I, **Chimusa ER*.** Dissecting genome-wide studies for microbiome-related metabolic diseases. Hum Mol Genet. 2020 Sep 30;29(R1):R73-R80. PMID: 32478833

b. Elsheikh SSM, **Chimusa ER**, Mulder NJ, Crimi A. Genome-Wide Association Study of Brain Connectivity Changes for Alzheimer's Disease. Sci Rep. 2020 Jan 29;10(1):1433. PubMed PMID: 31996736

c. Mufford M, Cheung J, Jahanshad N, van der Merwe C, Ding L, Groenewold N, Koen N, **Chimusa ER,** Dalvie S, Ramesar R, Knowles JA, Lochner C, Hibar DP, Paschou P, van den Heuvel OA, Medland SE, Scharf JM, Mathews CA, Thompson PM, Stein DJ. Concordance of genetic variation that increases risk for tourette syndrome and that influences its underlying neurocircuitry. Transl Psychiatry. 2019 Mar 22;9(1):120. PMID: 30902966

d. Elsheikh SSM, Bakas S, Mulder NJ, **Chimusa ER,** Davatzikos C, Crimi A. Multi-stage association analysis of glioblastoma gene expressions with texture and spatial patterns. Brainlesion. 2019;11383:239-250. PMID: 31482151

2. Genomics method development and Post Genome-wide Association Study

I contribute to designing a multi-population genetics Tools (Mugo et al., 2017). Genome-wide Association studies may miss disease genes with weak genetic effects or strong epistatic effects due to the single-marker testing approach commonly used. I addressed this challenge, by introducing a post GWAS tool "ancGWAS", an algebraic graph-based centrality measure that leverages the human protein-protein interaction information, local ancestry (in the case of admixed populations) and linkage disequilibrium data to mine GWAS summary statistics in deconvoluting the interactions between genes and identifying significant disease sub-networks underlying the pathogenesis of complex diseases (Chimusa et al., 2016). This tool holds potential for deconvoluting the interactions between genes underlying the diseases risk using GWAS summary statistics, estimating gene- and sub-network-specific ancestry and tests for case-control unusual difference in ancestry at the gene and sub-network level using the local-specific ancestry in admixed populations. I also contributed to develop a comprehensive and customized set of Gene Ontology (GO) based functional analysis tools that exploit the biological knowledge that GO offers in describing genes or groups of genes (Mazandu et al., 2015).

a. Mugo JW, Geza E, Defo J, Elsheikh SSM, Mazandu GK, Mulder NJ, **Chimusa ER*.** A multi-scenario genome-wide medical population genetics simulation framework. Bioinformatics. 2017 Oct 1;33(19):2995-3002. PMID: 28957497

b. Mazandu GK, **Chimusa ER*,** Mulder NJ. Gene Ontology semantic similarity tools: Survey on features and challenges for biological knowledge discovery. Brief Bioinform. 2017 Sep 1;18(5):886-901. PMID: 27473066

c. **Chimusa ER*,** Mbiyavanga M, Mazandu GK, Mulder NJ. ancGWAS: A post genome-wide association study method for interaction, pathway and ancestry analysis in homogeneous and admixed populations. Bioinformatics. 2016 Feb 15;32(4):549-556. PMID: 26508762

d. Mazandu GK, **Chimusa ER***, Mbiyavanga M, Mulder NJ. A-DaGO-Fun: An adaptable gene ontology semantic similarity-based functional analysis tool. Bioinformatics. 2016 Feb 1;32(3):477-479. PubMed PMID: 26476781

3. Detecting and characterizing population structure

Detecting and characterizing population structure is important for a wide range of applications, including correcting confounding due to population stratification and identifying signals of natural selection. I investigate the population genetics of African and South African Colored population (SAC), mixed ancestry population (de Wit et al., 2010). I constructed a panel of ancestry informative markers for the complex five-way admixed SAC population (Daya et al., 2013). I developed PROXYANC tool to account for inaccurate parental populations for admixed populations (Chimusa et al., 2013). Furthermore, I have significantly contributed to present a broad survey of polymorphisms in a novel array genotyping and copy number of polymorphism data set of self-identified ethnic/linguistic populations of Africa (Chimusa et al., 2015; Choudhury et al., 2017; Choudhury et al., 2020).

a. Choudhury A, Ramsay M, Hazelhurst S, Aron S, Bardien S, Botha G, **Chimusa ER**, Christoffels A, Gamieldien J, Sefid-Dashti MJ, Joubert F, Meintjes A, Mulder N, Ramesar R, Rees J, Scholtz K, Sengupta D, Soodyall H, Venter P, Warnich L, Pepper MS. Whole-genome sequencing for an enhanced understanding of genetic variation among South Africans. Nat Commun. 2017 Dec 12;8(1):2062. PMID: 29233967

b. Choudhury A, Aron S, Botigué LR, Sengupta D, Botha G, Bensellak T, Wells G, Kumuthini J, Shriner D, Fakim YJ, Ghoorah AW, Dareng E, Odia T, Falola O, Adebiyi E, Hazelhurst S, Mazandu G, Nyangiri OA, Mbiyavanga M, Benkahla A, Kassim SK, Mulder N, Adebamowo SN, **Chimusa ER**, Muzny D, Metcalf G, Gibbs RA; TrypanoGEN Research Group, Rotimi C, Ramsay M; H3Africa Consortium, Adeyemo AA, Lombard Z, Hanchard NA. High-depth African genomes inform human migration and health. Nature. 2020 Oct;586(7831):741-748. PMID: 33116287

c. **Chimusa ER***, Meintjies A, Tchanga M, Mulder N, Seoighe C, Soodyall H, Ramesar R. A genomic portrait of haplotype diversity and signatures of selection in indigenous southern African populations. PLoS Genet. 2015 Mar;11(3):e1005052. PMID: 25811879.

d. **Chimusa ER***, Daya M, Möller M, Ramesar R, Henn BM, van Helden PD, Mulder NJ, Hoal EG. Determining ancestry proportions in complex admixture scenarios in South Africa using a novel proxy ancestry selection method. PLoS One. 2013;8(9):e73971. PMID: 24066090.

e. de Wit E, Delport W, **Chimusa ER***, Meintjes A, Möller M, van Helden PD, Seoighe C, Hoal EG. Genome-wide analysis of the structure of the South African Coloured Population in the Western Cape. Hum Genet. 2010 Aug;128(2):145-53. PMID: 20490549

4. Dissecting Polygenic Risk Scores

I have made significant research contributions to dissecting genetics risk scores and highlight their research and clinical potential.

a. Damena D, **Chimusa ER***. Genome-wide heritability analysis of severe malaria resistance reveals evidence of polygenic inheritance. Hum Mol Genet. 2020 Jan 1;29(1):168-176. PMID: 31691794

b. Agamah FE, Mazandu GK, Hassan R, Bope CD, Thomford NE, Ghansah A, **Chimusa ER***. Computational/in silico methods in drug target and lead prediction. Brief Bioinform. 2019 Nov 10. PMID: 31711157

c. Awany D, Allali I, **Chimusa ER***. Tantalizing dilemma in risk prediction from disease scoring statistics. Brief Funct Genomics. 2018 Jul 22;18(4):211-219. PMID: 30605512

d. **Chimusa ER**, Zaitlen N, Daya M, Möller M, van Helden PD, Mulder NJ, Price AL, Hoal EG. Genome-wide association study of ancestry-specific TB risk in the South African Coloured population. Hum Mol Genet. 2014 Feb 1;23(3):796-809. PMID: 24057671

5. Data standards and ontology development

The last set of papers describe our activities in the development of data standards and ontology. The SCD Ontology is a significant contribution to the field and is being used for harmonizing SCD data and clinical decision making.

a. Kumuthini J, Mbiyavanga M, **Chimusa ER**, Pathak J, Somervuo P, Van Schaik RH, Dolzan V, Mizzi C, Kalideen K, Ramesar RS, Macek M, Patrinos GP, Squassina A. Minimum information required for a DMET experiment reporting. Pharmacogenomics. 2016 Sep;17(14):1533-45. PMID: 27548815

b. **Chimusa ER**, Mbiyavanga M, Masilela V, Kumuthini J. "Broadband" Bioinformatics Skills Transfer with the Knowledge Transfer Programme (KTP): Educational Model for Upliftment and Sustainable Development. PLoS Comput Biol. 2015 Nov;11(11):e1004512. PMID: 26583922

c. Sickle Cell Disease Ontology Working Group. The Sickle Cell Disease Ontology: enabling universal sickle cell-based knowledge representation. Database (Oxford). 2019 Jan 1;2019:baz118. doi: 10.1093/database/baz118. PMID: 31769834; PMCID: PMC6878945.

Complete List of Published Work on PubMed:

https://www.ncbi.nlm.nih.gov/myncbi/1r7CwVN_0wtAR/bibliography/public/?sortby=author&sdirection=ascending